

Daviti Gochitashvili

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EDUCATION

Binghamton University

PhD in Computational Condensed Matter Physics

Vestal, NY

Aug. 2020 – Dec. 2025 (Expected)

Free University of Tbilisi

BS in Physics

Tbilisi, Georgia

Sep. 2015 – Aug. 2019

FINANCIAL INDUSTRY EXPERIENCE

Bank of Georgia, Analyst – Developer, *Tbilisi, Georgia*

Jul. 2019 – Aug. 2020

- Developed API web services for the Core Banking System using Java and PL/SQL, enhancing system responsiveness, streamlining internal workflows, and enabling efficient handling of high-volume banking transactions.
- Designed and implemented quantitative finance and banking-related functions for the Cash Operations and Liens Agile (Scrum) development squad, enabling data-driven decision-making and faster project delivery.
- Collaborated with Revenue Service officials to gather requirements, analyze complex banking rules, ensure compliance, and convert them into actionable system functions and automated solutions that improved efficiency.
- Accelerated cash counting operations by ~ 5 times and automated money counting machines to generate detailed operational and regulatory reports, leveraging complex SQL queries for analytics and process validation.

ACADEMIC EXPERIENCE

Binghamton University, Graduate Researcher/Associate

Sep. 2020 – Present

My publications are available at [Google Scholar](https://scholar.google.com/citations?user=datogochitashvili)

- Contributed to the development of the open-source MAISE package, written in C and Python, which streamlines the construction, testing, and application of accurate machine learning models for describing atomic interactions.
- Utilized the accurate neural network interatomic potentials for discovering new stable compounds using evolutionary algorithm. Predicted 29 novel viable stable metal alloys at different temperatures and pressures.
- Investigated the stability of ambient-pressure superconductors using Density Functional Theory. Developed phase diagrams and proposed a thermodynamic pathway for synthesizing new layered metal borocarbides.
- Implemented hyperdimensional optimization to enhance the efficiency of structure search and proposed an atom block seeding method that significantly improved material prediction for periodic and non-periodic systems.

Binghamton University, Teaching Assistant

Sep. 2020 – Present

- Led two discussion sections of 30 students each for general physics courses. Provided individual student support.
- Conducted lab demonstrations, set up equipment for experiments, assisted students during lab course time.

SKILLS

Technical skills:	Machine learning, data analysis & visualization, statistics, stochastic calculus.
Programming languages:	Python (NumPy, pandas, SciPy, matplotlib, scikit-learn), PL/SQL, Java, C++.
Software:	Git, Bash, Linux, Slurm, MATLAB, Wolfram Mathematica, Microsoft Excel.
Languages:	Georgian (native), English (fluent), Russian (conversational), German (basic).

HONORS AND FELLOWSHIPS

Provost's Doctoral Summer Fellowship (Binghamton University).

2020 – 2024

Knowledge Fund Scholarship (Free University of Tbilisi).

2015 – 2019